

Subnetting

Subnetting is done to divide a large network into smaller, manageable networks (subnets). It's a practical concept used within the **TCP/IP Model**

Here are the main reasons:

1. Efficient Use of IP Addresses

Without subnetting, a network might waste a lot of IP addresses. Subnetting lets you **divide one large network into smaller ones**, so addresses are used more efficiently allocating only the required number of IPs to each subnet

2. Reduces Network Traffic (Less Congestion)

Each subnet is a separate broadcast domain which limits broadcast messages thus increasing performance.

3. Improves Security

Each subnet can act like a **separate mini-network**, so:

- You can **restrict access** between subnets
- Sensitive departments (e.g., finance, HR) can be isolated and controlled via routers.

4. Better Network Management

It becomes easier to troubleshoot problems, monitor network activity and organize devices based on logical grouping (e.g., HR, IT, Sales)

5. Supports Scalability

Subnetting allows a network to **grow in an organized way** without becoming messy or inefficient. It also enables efficient routing (used with longest prefix match)

VLAN and Its Assignment Types

Definition of VLAN

A **Virtual Local Area Network (VLAN)** is a logical grouping of devices within a network that allows them to communicate as if they were on the same physical LAN, regardless of their physical location. VLANs improve network management, security, and performance by separating broadcast domains.

VLAN Assignment Types

1. Interface-Based VLAN (Static VLAN)

- Devices are assigned to a VLAN based on the switch port they connect to.
- Most commonly used VLAN assignment method.
- Easy to configure and manage.

2. MAC Address-Based VLAN

- Devices are assigned to VLANs according to their MAC addresses.
- A device remains in the same VLAN even if it moves to a different switch port.

3. Protocol-Based VLAN

- VLAN membership is determined by the network protocol being used (such as IP or IPX).
- Different protocols can be assigned to different VLANs.

4. IP Subnet-Based VLAN

- Devices are assigned to VLANs based on their IP addresses or subnet membership.
- Useful for organizing users according to network segments.

5. Policy-Based VLAN

- VLAN assignment is based on predefined policies such as user identity, application type, or security requirements.
- Provides flexible and dynamic VLAN management.

Steps Involved in VLAN Assignment

1. Analyze the network requirements and identify departments, user groups, or services that need to be separated.
2. Create VLANs on the switch by configuring unique VLAN IDs and names.
3. Assign switch ports or devices to the appropriate VLANs based on ports, MAC addresses, IP subnets, or policies.
4. Configure trunk links between switches to allow traffic from multiple VLANs to pass through a single physical connection.
5. Configure inter-VLAN routing using a router or Layer 3 switch if communication between different VLANs is required.
6. Verify VLAN creation and port assignments using commands such as **display vlan** or **show vlan**. Test connectivity within the same VLAN and between VLANs (if routing is enabled).

VLAN Implementation via Router's Physical Interfaces

In this method, each VLAN is connected to a separate **physical interface** on the router. The router's Layer 3 interfaces act as **gateways** and forward traffic between different network segments. Since router physical interfaces cannot process VLAN-tagged frames, the switch ports connected to the router must be configured as Access ports.

Each router interface can serve as the gateway for **only one VLAN**. Therefore, the number of router interfaces required increases with the number of VLANs.

Disadvantages

- Requires one physical router interface per VLAN.
- Routers usually have a limited number of physical interfaces.
- Poor scalability for networks with many VLANs.
- Higher hardware requirements and costs.

VLAN Implementation Using a Router's Sub-Interfaces (Router-on-a-Stick)

A sub-interface is a logical interface created on a router's physical Ethernet interface. It is identified by a **physical interface number and a sub-interface number** (for example, G0/0/1.10).

Like a physical interface, a sub-interface can perform Layer 3 forwarding. Unlike a physical interface, it can also process VLAN-tagged frames.

Multiple sub-interfaces can be created on a single physical interface, with each sub-interface serving as the gateway for a different VLAN. The router's physical interface is connected to a **Trunk port** on the switch, allowing traffic from multiple VLANs to pass through a single link. The router then uses sub-interfaces to route traffic between VLANs.

Advantages

- One physical interface can support multiple VLANs.
- Requires fewer router interfaces.
- Better scalability than using separate physical interfaces.
- Cost-effective solution for inter-VLAN routing.

Layer 3 Switch and VLANIF Interfaces

A **Layer 3 switch** combines the functions of both a **Layer 2 switch** and a **router**. In addition to Layer 2 switching, it can perform routing using **VLANIF interfaces**.

A **VLANIF (VLAN Interface)** is a logical Layer 3 interface associated with a VLAN. It acts as the **gateway** for devices within that VLAN and can process VLAN-tagged traffic, enabling communication between different VLANs.

Each VLANIF interface corresponds to a specific VLAN, and its interface number is the same as the VLAN ID. For example, **VLANIF 10** corresponds to **VLAN 10**.

When devices in different VLANs need to communicate, traffic is sent to their respective VLANIF interfaces, and the Layer 3 switch routes the traffic between VLANs.

Advantages

- No external router is required.
- Faster inter-VLAN communication because routing is performed within the switch.
- Better scalability and performance.
- Simplified network design.

Eth-Trunk (Link Aggregation)

1. Network Reliability

Network reliability refers to the capability of ensuring nonstop network services when single or multiple points of failure occur on a device or link. It can be implemented at the card, device, and link levels.

2. Eth-Trunk (Link Aggregation)

Ethernet link aggregation, also called Eth-Trunk, bundles multiple physical links into a logical link to increase link bandwidth, without having to upgrade hardware.

Benefits

- Increases bandwidth.
- Provides link redundancy.
- Supports load balancing.
- Improves network reliability.
- Eliminates STP blocking on aggregated links.

Key Terms

Term	Meaning
Eth-Trunk	Logical interface formed by multiple links
Member Interface	Physical interface inside the trunk
Active Interface	Interface forwarding traffic
Inactive Interface	Backup interface
Max Active	Maximum active interfaces allowed
Min Active	Minimum interfaces required for trunk operation

Based on whether the Link Aggregation Control Protocol (LACP) is enabled, link aggregation can be classified into manual mode and LACP mode.

3. Manual Mode (Static Eth-Trunk)

In Manual Mode, the administrator manually creates the Eth-Trunk and adds member interfaces. LACP is not used, so no negotiation occurs between devices. All member links are typically in forwarding state unless they fail physically.

Advantages

- Easy to configure.
- Works with older devices.

Defects (Important Exam Question)

1. In manual mode, no protocol messages are exchanged, so incorrect peer connections cannot be detected, which may cause traffic issues even if the trunk is UP.
2. Only physical link failures are detected. Logical failures (traffic not passing despite link UP) remain undetected.

Manual Mode = All links active + No negotiation + Cannot detect wrong peer + Cannot detect logical failures.

4. LACP Mode (Dynamic Eth-Trunk)

A link aggregation mode that uses the LACP protocol. Devices exchange Link Aggregation Control Protocol Data Units (LACPDU) to ensure that the peer interfaces are member interfaces that belong to the same Eth-Trunk and are on the same device.

An LACPDU contains the device priority, MAC address, interface priority, and interface number.

Functions

- Automatic negotiation.
- Detects wrong peer connections.
- Detects logical and physical failures.
- Supports active and backup links.
- Provides automatic failover.

Actor Election

The Actor is the device that controls LACP decisions and selects active interfaces.

Election Rules:

1. Lower System Priority wins.
2. If equal, lower MAC Address wins.

The other device becomes the **Partner**.

Active Interface Selection

The Actor selects active links using:

1. Lower Interface Priority has higher preference.
2. If equal, lower Interface Number is selected first.
3. Selection is limited by the configured **max active-link number**.

Remaining interfaces are placed in **backup (inactive) state**.

Active/Backup Mechanism

- Active links forward traffic and share load.
- Backup links stay standby.
- If an active link fails, a backup link automatically becomes active.
- No manual intervention is required.

Example:

3 links, max-active = 2

- GE0/0/1 → Active
- GE0/0/2 → Active
- GE0/0/3 → Backup

If GE0/0/1 fails → GE0/0/3 becomes active automatically.

5. Load Balancing

Eth-Trunk distributes traffic across active links.

Per-Packet

Traffic is sent packet by packet across different links. This can cause out-of-order delivery, so it is not recommended.

Per-Flow (Recommended)

Traffic of the same flow is always sent through the same link. This maintains packet order and provides stable load balancing across links.

STP — Complete Exam Notes

1. Why STP Exists: The Problem

Redundant links can create Layer 2 loops because Ethernet frames do not have a TTL field. So the following consequences can occur.

- Broadcast, unknown unicast, and multicast (BUM) frames may circulate indefinitely, causing broadcast storms and network congestion.
- The same MAC address may appear on different ports, leading to MAC address flapping.
- Frequent MAC table updates create instability and increase switch CPU utilization.
- Excessive looping traffic can disrupt normal forwarding and potentially bring the network down

2. What STP Does

STP (IEEE 802.1D) runs on switches, exchanges BPDUs, elects a root bridge, calculates a loop-free tree topology, and places specific ports into a blocking state to prevent loops. If a link fails, STP recalculates the topology and activates alternate paths.

3. Key Concepts (Building Blocks)

BID (Bridge ID) — Consists of 16-bit priority + 48-bit MAC address. Uniquely identifies each switch. Smaller BID = higher priority in election.

Root Bridge — Switch with smallest BID. Serves as the reference point for all path calculations and is only one per network. All ports on root bridge are Designated ports.

Cost — Per-port value based on link bandwidth. Higher bandwidth = lower cost.

- 1 Gbps → cost 4 (802.1D) / 20,000 (802.1t)
- 100 Mbps → cost 19 / 200,000

RPC (Root Path Cost) — Sum of all inbound port costs along path from switch to root bridge. Lower RPC is preferred.

PID (Port ID) — Consists of 4-bit priority + 12-bit port number. Default priority is 128. Used as tiebreaker when RPC and BID are equal.

BPDU (Bridge Protocol Data Unit): Control messages exchanged between switches to build and maintain the STP topology. Two types:

- **Configuration BPDU**: carry topology information for calculation and are sent periodically every Hello time.

- **TCN BPDU** — Topology Change Notification, notify switch when topology changes

4. BPDU Comparison Rules (Election Tiebreakers — in order)

1. Smallest Root BID → elect Root Bridge
 2. Smallest RPC → elect Root Port
 3. Smallest sender BID → tiebreaker
 4. Smallest PID of receiver interface → final tiebreaker
-

5. STP Election Process

Step 1 — Root Bridge Election

All switches initially advertise themselves as the root bridge by sending BPDUs. The switch with the lowest BID is elected as the root bridge.

Step 2 — Root Port Election

Each non-root switch selects one Root Port (RP), which is the port with the lowest Root Path Cost (RPC) to the root bridge.

Step 3 — Designated Port Election

For each network segment, one Designated Port (DP) is elected. It is the port that advertises the best BPDU toward that segment. All ports on the root bridge are Designated Ports.

Step 4 — Blocking Non-Designated Ports

Any port that is neither a Root Port nor a Designated Port is placed in a blocking state to prevent Layer 2 loops.

6. STP Timers

Timer	Default	Purpose
Hello Time	2s	Interval at which the root bridge sends Configuration BPDUs
Forward Delay	15s	Time spent in Listening state AND in Learning state (15s each)
Max Age	20s	Max time switch waits without BPDU before declaring link dead
Convergence Time	50s	Worst-case time for STP to reconverge after a topology change, calculated as Max Age (20s) + Listening (15s) + Learning (15s)

7. STP Port States (5 States)

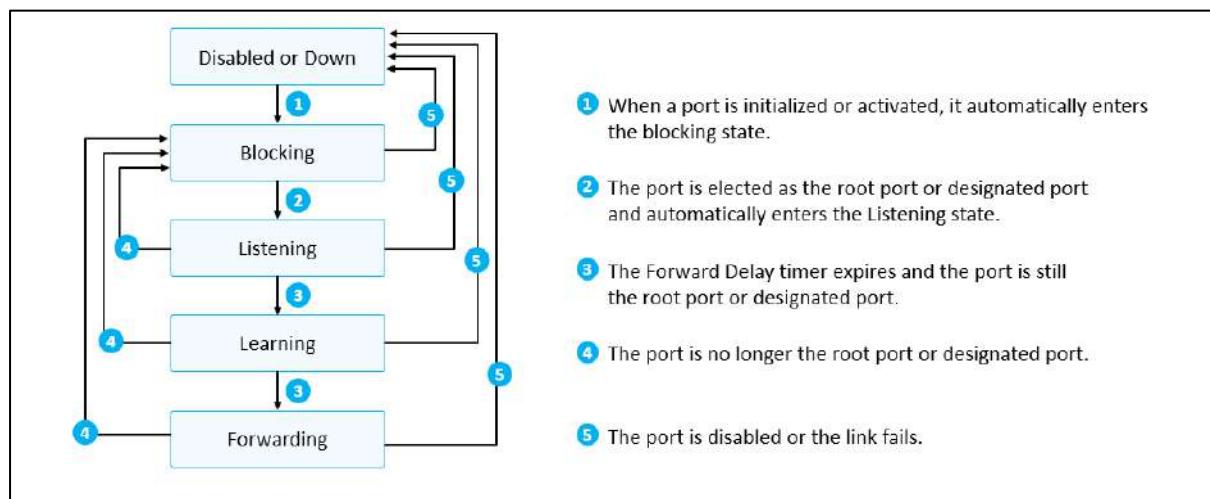
State	Send BPDU	Receive BPDU	Learn MAC	Forward Data
Disabled	No	No	No	No
Blocking	No	Yes	No	No

Listening	Yes	Yes	No	No
Learning	Yes	Yes	Yes	No
Forwarding	Yes	Yes	Yes	Yes

Only **Root Port** and **Designated Port** reach Forwarding state.

Port State	Description
Disabled	The port cannot send or receive BPDUs or service data frames. That is, the port is Down.
Blocking	The port is blocked by STP. A blocked port cannot send BPDUs but listens to BPDUs. In addition, the blocked port cannot send or receive service data frames or learn MAC addresses.
Listening	STP considers the port in Listening state as the root port or designated port, but the port is still in the STP calculation process. In this case, the port can send and receive BPDUs but cannot send or receive service data frames or learn MAC addresses.
Learning	A port in Learning state listens to service data frames but cannot forward them. After receiving service data frames, the port learns MAC addresses.
Forwarding	A port in Forwarding state can send and receive service data frames and process BPDUs. Only the root port or designated port can enter the Forwarding state.

8. STP Port State Transitions



If port loses RP/DP role at any stage → back to **Blocking**. If link fails or port disabled → **Disabled**.

9. Topology Change & Convergence Times

Root Bridge Failure:

- Switches wait **Max Age (20s)** for BPDU to expire
- Re-elect root bridge
- New port transitions through Listening (15s) + Learning (15s)
- **Total ≈ 50s** to recover

Direct Link Failure:

- SW detects link down immediately
- Alternate port transitions Listening (15s) + Learning (15s)
- **Total ≈ 30s** to recover

Indirect Link Failure:

- SW waits **Max Age (20s)** first, then 30s transition
- **Total ≈ 50s** to recover

TCN (Topology Change Notification) process: When topology changes, switch sends TCN BPDU upstream → root bridge sets TC bit in Config BPDUs → all switches shorten MAC aging time → stale MAC entries cleared faster.

10. STP Disadvantages

- Convergence is slow, typically taking 30–50 seconds to recover after a failure, which is unsuitable for modern networks.
 - Port states (blocking, listening, learning) do not forward traffic, making topology understanding and troubleshooting less intuitive.
 - The protocol is timer-based, so it may wait for timers to expire even when a failure has already been detected.
 - BPDUs must propagate hop-by-hop through switches, which can slow convergence in larger networks.
 - A single spanning tree is used for all VLANs, preventing load balancing and leaving some links underutilized.
-

11. RSTP — Improvements Over STP (IEEE 802.1w)

RSTP is an improved version of STP that provides faster convergence and better failure recovery, reducing downtime from ~50 seconds in STP to a few seconds. It replaces timer-heavy behavior with a rapid handshake mechanism.

New Port Roles (4 total):

- **Root Port (R)** — Best path toward the root bridge (same as STP)
- **Designated Port (D)** — Best port on a LAN segment (same as STP)
- **Alternate Port (A)** — Backup path to the root bridge; immediately becomes RP if the current RP fails
- **Backup Port (B)** — Backup for a Designated Port on the same shared segment (rare in modern switched networks)

Edge Port: Edge Ports are connected to end devices and transition directly to the forwarding state without delay. This is similar to STP PortFast, and they lose edge status immediately if a BPDU is received.

Reduced Port States (3 states only):

RSTP State	Replaces STP State(s)
Discarding	Disabled + Blocking + Listening
Learning	Learning
Forwarding	Forwarding

BPDU improvements:

- BPDUs are generated by all switches instead of only the root bridge, allowing faster failure detection and improved synchronization.
- Failure detection is faster because BPDU timeout is based on $3 \times \text{Hello Time}$ (about 6 seconds) instead of Max Age (20 seconds).
- BPDU includes extended flags to carry port role and state information for synchronization.
- Topology changes are handled more efficiently by directly flushing MAC entries on affected ports, avoiding slow TCN propagation used in STP.

OSPF — Complete Exam Notes

1. Why Dynamic Routing Protocols?

Static routes are manually configured and are best suited for small or stable networks. They have several limitations:

- Poor scalability: As the number of routers and networks increases, manual configuration and maintenance become complex and time-consuming.
- No automatic adaptation: Static routes do not adjust to network changes, so any topology change requires manual reconfiguration.
- No automatic recovery: If a link fails, routes must be manually updated, causing downtime unless a backup route exists.

Dynamic routing protocols solve these problems by allowing routers to automatically learn available paths, and exchange/update routing information when the network topology changes without manual intervention.

2. Classification of Dynamic Routing Protocols

By Scope (AS-based):

IGP (Interior Gateway Protocol) — runs *within* one Autonomous System (AS). Examples: RIP, OSPF, IS-IS

EGP (Exterior Gateway Protocol) — runs *between* different Autonomous Systems.
Example: BGP

By Working Mechanism/Algorithm:

Distance Vector (e.g. RIP):

- Routers send their routing tables to directly connected neighbors periodically
- Routers learn routes indirectly through neighbors, not full network topology like where and how far the destination is.
- Simple but slow convergence and can form routing loops

Link State (e.g. OSPF, IS-IS):

- Routers exchange Link-State Advertisements (LSAs), not full routing tables. LSAs describe directly connected links and their costs
- Each router builds a complete Link-State Database (LSDB) of the network
- Each router runs the SPF (Dijkstra) algorithm to compute a shortest path tree with itself as the root and builds its routing table from it
- It is fast convergence, scalable, and loop-free

Basic Link-State process:

1. Form neighbor adjacencies
2. Flood LSAs and build LSDB
3. Run SPF (Dijkstra algorithm)
4. Install best routes in routing table

3. Introduction to OSPF

OSPF (Open Shortest Path First) is a link-state routing protocol widely used in enterprise and campus networks. Its characteristics are:

- It is an open standard protocol (not vendor-specific).
- OSPF uses Link-State Advertisements (LSAs) to share routing information within an area.
- Routers build a Link-State Database (LSDB), which represents the network topology.
- Each router runs the SPF (Dijkstra) algorithm to compute a shortest path tree with itself as the root and builds its routing table from it.
- Only the best routes are installed into the routing table.
- OSPF supports VLSM and route summarization.
- It uses a multi-area design to improve scalability in large networks.

4. OSPF Packet Types (5 Types)

Term	Meaning
------	---------

Hello	Protocol to discover OSPF neighbors and confirm reachability. It is also used in DR election
LSA	Information a router sends/receives about network reachability; builds the LSDB
LSU (Link State Update)	Packet that carries LSAs
LSR (Link State Request)	Used by router to request specific LSA info from neighbor
LSAck	Router confirms it received an LSU
DD / DBD (Database Description)	Carries LSDB summary (headers only) during sync; not full contents
LSDB	Database storing all LSAs = full topology map

5. Basic OSPF Concepts

Area

Logical group of OSPF routers identified by an **Area ID**. All multi-area OSPF networks **must** have Area 0 (backbone). LSA flooding is contained within an area. ABRs connect areas to backbone.

Router ID

32-bit unique identifier for each OSPF router (looks like IP, e.g. 1.1.1.1).

Router ID selection (automatic, in order):

1. Manually configured router-id (preferred, most stable)
2. Highest IP address on any **Loopback** interface
3. Highest IP address on any **active physical** interface

Three OSPF Tables

- **Neighbor Table (Adjacency Database)** — lists all discovered OSPF neighbors and tracks their adjacency state (e.g., Down, Init, 2-Way, Full).
- **Link-State Database (LSDB)** — stores all Link-State Advertisements (LSAs) received from routers in the same area; it represents the complete topology of that area.
- **OSPF Routing Table (Shortest Path First / SPF Tree output)** — contains the best routes calculated from the LSDB using the SPF algorithm, including destination networks, cost (metric), and next-hop information.

6. Neighborhood vs Adjacency

Neighbors are routers that:

- Reside on the same network segment.
- Exchange Hello messages (via multicast 224.0.0.5)

Adjacencies are routers that:

- Are neighbors AND
- Have exchanged **LSUs and DBD packets** and have fully synchronized their LSDBs

Not all neighbors become adjacent. On broadcast/NBMA networks, DROthers only reach 2-Way with each other (not Full).

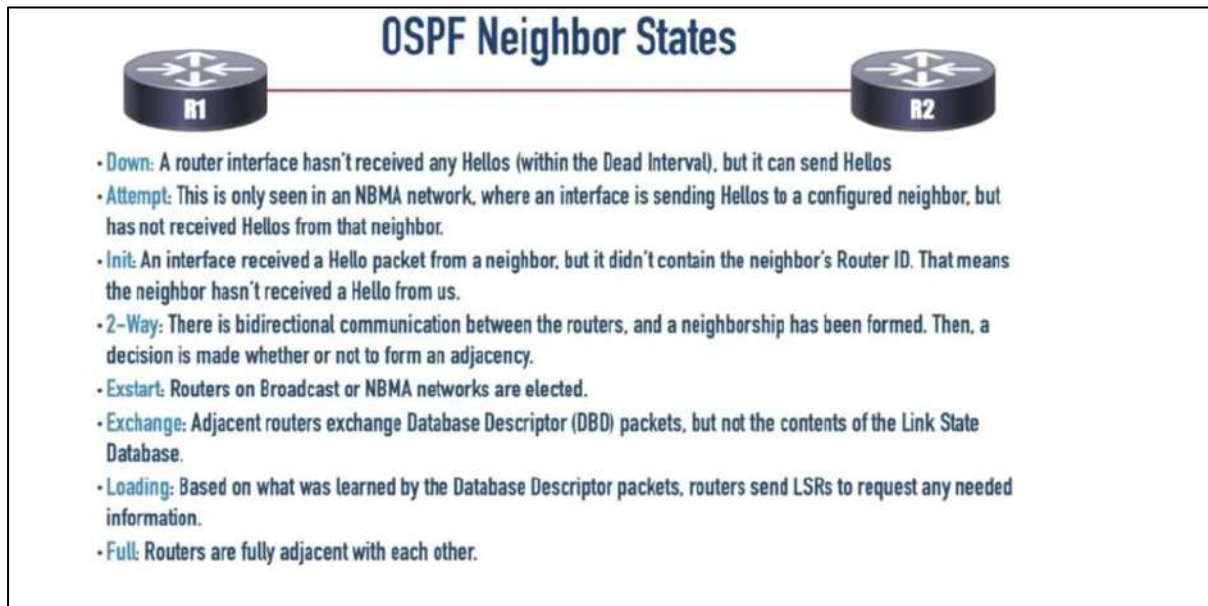
Both routers **must match** on all of these for neighbor relationship to form:

- Matching Area ID
- Matching Authentication (type and password)
- Matching Subnet (same network segment)
- Matching Timers (Hello and Dead intervals)
- Matching Stub Flags (area type must agree)
- Matching MTU (checked at ExStart/Exchange — mismatch blocks adjacency here)

7. Adjacency Establishment Process (5 Steps)

- **Hello Exchange (Neighbor Discovery)** — Routers send Hello packets to discover neighbors, match parameters (area ID, timers, network type, authentication). If matched, they move through Down → Init → 2-Way states.
- **ExStart** — Routers establish the master/slave relationship (based on highest Router ID) and Master decide sequence numbers for Database Description (DBD) exchange.
- **Exchange** — Routers exchange DBD (Database Description) packets, which contain LSA summaries (headers only), not full LSAs.
- **Loading** — Routers request missing or outdated LSAs using LSR (Link-State Request) which is received via LSU (Link-State Update) and acknowledged using LSAck.
- **Full** — LSDBs are fully synchronized with routers becoming fully adjacent and SPF runs to build the routing table.

8. OSPF Neighbor States (8 States)



Stable states: **2-Way and Full**

OSPF Timers

Hello Timer — interval router sends Hello packets out of OSPF-enabled interface.

Dead Timer — time interface waits without receiving a Hello before declaring neighbor down.

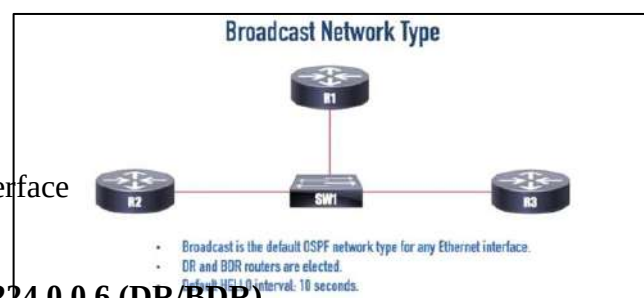
$$\text{Dead Timer} = \text{Hello Timer} \times 4$$

Timers must match between neighbors (mismatch = no neighborship).

8. OSPF Network Types (4 Types)

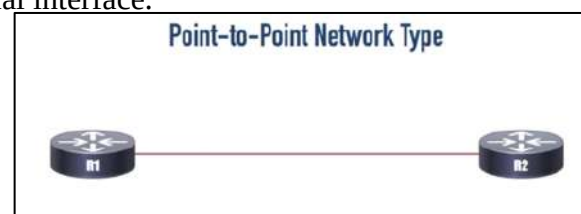
Broadcast (BMA)

- Default OSPF network type for any Ethernet interface
- Supports multiple devices and broadcasts
- DR/BDR election is performed
- Uses multicast **224.0.0.5 (all SPF routers)** and **224.0.0.6 (DR/BDR)**
- Default timers: Hello 10s, Dead 40s



Point-to-Point (P2P)

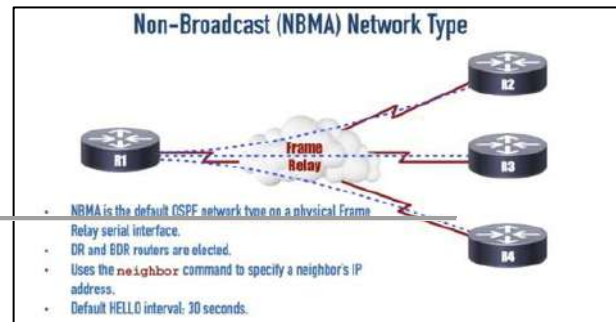
- Default OSPF network type on a non-Frame Relay serial interface.



- Exactly two routers on the link
- No DR/BDR election
- Routers form full adjacency directly
- Default timers: Hello 10s, Dead 40s

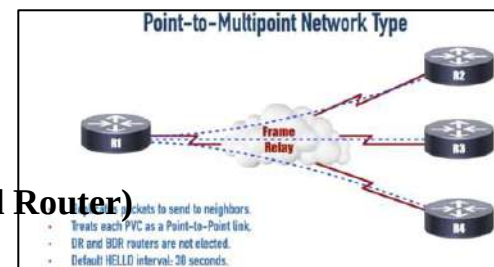
Non-Broadcast Multi-Access (NBMA)

- Default network type on a physical Frame Relay serial interface
- Supports multiple routers, but no broadcast.
- DR/BDR election is performed
- Neighbors must be manually configured
- Default timers: Hello 30s, Dead 120s



Point-to-Multipoint (P2MP)

- Treats each neighbor as a separate point-to-point link and replicates packets to send to neighbors.
- No DR/BDR election
- Commonly used to simplify partial-mesh NBMA networks
- Default timers: Hello 30s, Dead 120s



9. DR and BDR (Designated Router / Backup Designated Router)

Problem on MA (multi-access) networks: If every OSPF router forms adjacency with every other router, on a network of N routers there are $N(N-1)/2$ adjacencies → massive BPDU/LSA flooding → wastes bandwidth and device resources.

Solution: Elect a **DR** and **BDR** on MA networks (Broadcast and NBMA types).

Three router roles on MA network:

- **DR (Designated Router)** — central hub; all others form adjacency with it. Collects and distributes LSAs for the segment.
- **BDR (Backup DR)** — monitors DR; takes over instantly if DR fails. Also forms full adjacency with all routers.
- **DRother** — all remaining routers. Form Full adjacency only with DR and BDR. With each other: only **2-Way** (no adjacency).

DR Election Rules:

1. Highest **OSPF interface priority** wins (default = 1, range 0–255; priority 0 = never elected DR/BDR)
2. Tiebreak: highest **Router ID**

DR/BDR election is **non-preemptive** — once elected, DR keeps role even if a higher-priority router joins later (unless DR fails).

LSA flooding on MA: DR Others send LSAs to **224.0.0.6** (DR/BDR multicast). DR redistributes to all via **224.0.0.5** (all OSPF routers multicast).

10. OSPF Router Types (Multi-Area)

Router Type	Description
Internal Router (IR)	All interfaces in same single area
Backbone Router (BR)	Has at least one interface in Area 0 (backbone)
ABR (Area Border Router)	Connects two or more areas and has interfaces in multiple areas. Maintains separate LSDB per area. Summarizes routes between areas.
ASBR (Autonomous System Boundary Router)	A router within an OSPF system that connects to an external routing domain and redistributes external routes into OSPF.

One router can have multiple roles simultaneously (e.g. ABR + BR if it connects Area 1 to Area 0).

11. Single Area vs Multi-Area OSPF

Single Area problems (as network grows):

- LSDB grows huge → high memory/CPU usage on every router
- Any topology change triggers SPF recalculation on entire network
- LSA flooding affects all routers everywhere
- Routing table grows massive

Multi-Area solution:

- LSA flooding contained within each area
- SPF recalculation only affects routers in the changed area
- Routes summarized at ABR → smaller routing tables in other areas
- Better scalability for large enterprise networks
- **Area 0 (backbone)** is mandatory; all other areas must connect to Area 0 (directly or via virtual link)

Virtual Links: If an area cannot connect directly to **Area 0**, a virtual link is created through a transit area to provide a logical connection. OSPF routing updates pass through the virtual link, while data traffic follows normal routing paths without tunneling overhead.

Circuit Switching vs Packet Switching

Circuit Switching

Definition

Circuit switching is a communication method in which a dedicated, exclusive physical path is established between the sender and receiver before any data transmission begins. This reserved path remains allocated for the entire duration of the session and is released only when the connection is terminated.

How it works

The process occurs in three phases:

- **Connection setup** — the network reserves a fixed-bandwidth channel end-to-end.
- **Data transfer** — voice or data flows continuously along the reserved path with constant latency.
- **Connection teardown** — once the session ends, the dedicated resources are released back to the network.

Key characteristics

- The same bandwidth is reserved throughout the session, regardless of activity.
- Low and predictable latency is maintained during communication.
- Bandwidth is wasted during idle periods, such as pauses in a phone call.
- Data always travels along the same pre-established route.

Example:

A traditional telephone call uses circuit switching. Once the call is connected, a dedicated communication channel is reserved for both users until the call ends, even during periods of silence.

Limitation

The primary drawback of circuit switching is resource inefficiency. Once a circuit is established, no other communication can use that bandwidth — even during periods of complete silence. As voice traffic grew exponentially and IP-based data became dominant, this model proved economically and technically unsustainable, driving the transition to packet switching.

Packet Switching

Definition

Packet switching is a communication method in which data is broken into small, independent units called packets. Each packet carries its own header containing source address, destination

address, and sequence number. Packets are routed independently across the network and may take different paths, before being reassembled in the correct order at the destination.

How it works

Packet switching operates without any pre-established connection:

- **Segmentation** — the sending device breaks data into fixed-size packets.
- **Independent routing** — each router along the network reads the packet header and forwards it along the best available path using dynamic routing protocols (e.g. OSPF, BGP).
- **Statistical multiplexing** — multiple users share the same physical link simultaneously thus improving efficiency.
- **Reassembly** — the destination device reorders all received packets using sequence numbers to reconstruct the original data.

Key characteristics

- Bandwidth is shared dynamically, allowing available capacity to be used by other traffic.
- Packet delays may vary depending on network load and routing paths.
- Packets can be automatically re-routed if a network link fails.
- The network can support a very large number of simultaneous communications.

Example:

When browsing a website, the webpage data is broken into packets and sent across the Internet. Different packets may travel through different routes before reaching your device.

Comparison Table (Long Answer — Must Know)

Feature	Circuit Switching	Packet Switching
Path	Dedicated, pre-established	Dynamic, per-packet routing
Resource Reservation	Fixed — reserved throughout call	Shared — used only when needed
Efficiency	Low (idle resources wasted)	High (statistical multiplexing)
Delay	Constant, predictable	Variable (queuing delay possible)
Reliability	Path failure = total failure	Resilient — alternate routes
Overhead	Low (no per-packet addressing once connected)	Higher (each packet has header)

QoS	Guaranteed, consistent	Best-effort (QoS mechanisms needed)
Setup Required	Yes — connection must be established first	No — send data immediately
Example	Traditional telephone (PSTN)	Internet (IP networks), TCP/IP
Cost	Expensive (dedicated resources)	Cost-effective (shared resources)

Two Types of Packet Switching

- **Datagram Packet Switching:** Each packet routed independently. May arrive out of order. Used in Internet/IP. Connectionless.
- **Virtual Circuit Packet Switching:** Path is negotiated before transfer (like a logical circuit) but resources not strictly reserved. Packets follow same path. Examples: Frame Relay, ATM, MPLS.